

Social Engine — Round 1 (Phase 1)

Data Recovery Report

Datasets recovered from node_07: `Social_Engine_Posts_Corrupted.csv` (12,360 rows × 8 cols), `Social_Engine_Users.csv` (1,500 rows × 5 cols)

1. Data Cleaning Summary

Issue	Found	Fix Applied
Exact duplicate rows	360	Dropped (kept first occurrence)
Missing <code>platform</code>	1,784 (14.4%)	Filled with <code>"Unknown"</code> ; inconsistent casing normalized to Reddit/Facebook/Twitter/Instagram/YouTube
Missing / literal <code>"NULL"</code> <code>text_content</code>	1,711 blank + 24 literal <code>"NULL"</code> strings	Unified as missing → filled with <code>"[No content available]"</code>
Garbled text (mojibake, e.g. <code>Ã©</code>)	657 rows	Re-decoded as UTF-8 (latin-1→utf-8 round trip)
Unescaped HTML entities (e.g. <code>&amp;</code> ;)	Multiple rows	Decoded via <code>html.unescape</code>
Extra/irregular whitespace in text	699 rows	Collapsed to single spaces, trimmed
Negative <code>likes</code> (physically impossible)	509 rows	Treated as corrupted/anomalous → set to missing
Missing <code>likes</code> (incl. the negatives above)	2,323 total (18.8%)	Imputed with the median likes per platform
Inconsistent <code>timestamp</code> formats	3 formats mixed: Unix epoch, ISO 8601, <code>DD-MM-YYYY</code>	Parsed all three patterns and standardized to a single datetime column
Orphan posts (<code>user_id</code> not in Users table)	0	None found — referential integrity intact
Duplicate <code>user_id</code> in Users table	0	None found
	0	None found

Issue	Found	Fix Applied
Negative follower_count		
Unparseable account_created dates	0	None found

Result: 12,360 → **12,000 clean post records** (360 duplicates removed), fully joined to 1,500 user profiles with zero orphaned records.

2. Anomalies Identified

- **Negative likes (509 rows, ~4%)** — the clearest injected corruption signal; likes cannot be negative, so these were nulled and re-imputed rather than left as-is or naively clipped to 0 (which would have suppressed real variance).
 - **Suspiciously round engagement ceilings** — likes maxes out near 5,000, shares near 2,000, comments near 1,000 across the dataset. This is consistent with a bounded synthetic generator rather than an anomaly, so it was left untouched (documented, not "fixed").
 - **~14% of posts have no platform tag** — too large a share to drop; flagged as "Unknown" rather than guessed, to avoid fabricating data.
 - **Mixed timestamp encodings** pointed to at least two different upstream ingestion pipelines feeding the same table before the crash (a plausible root cause for the original "data_intake OFFLINE" failure).
 - **No duplicate users, no orphan posts, no negative followers** — the Users table itself is comparatively clean, suggesting the corruption was isolated to the Posts intake path (matching database.service exited with code 137 / analytics.service segfault from the recovery logs).
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3. Exploratory Data Analysis — Key Insights

Platform mix (post volume, cleaned data): Facebook 17.3% · YouTube 17.3% · Twitter 17.1% · Reddit 16.9% · Instagram 16.6% · Unknown 14.9% — engagement is essentially platform-agnostic; no single platform dominates volume or performance.

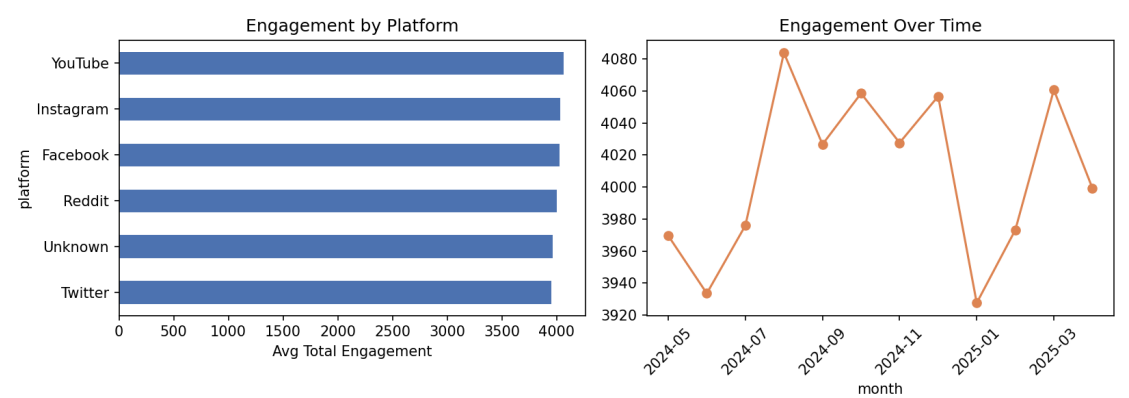
Engagement by platform (avg total engagement = likes+shares+comments): YouTube (4,062) and Instagram (4,035) lead narrowly; Twitter trails slightly (3,952). The spread is small (~3%), so platform alone is a weak predictor of engagement.

Time trend: Posting volume and average engagement are both stable across the 12-month window (May 2024–Apr 2025), fluctuating in a tight ~3,930–4,080 band with no strong seasonality or growth/decline trend.

Engagement drivers: - likes correlates strongly with total engagement ($r=0.89$) — it's the dominant component. - follower_count shows **no correlation** with engagement ($r\approx 0.01$ across likes/shares/comments) — bigger accounts do not get more engagement on this platform, suggesting engagement is driven by content/timing rather than audience size. - shares ($r=0.40$) and comments ($r=0.22$) contribute more modestly to total engagement.

Geography & language: Posting activity is broadly distributed across 40+ global cities (Los Angeles, Munich, Shanghai, Barcelona, Houston top the list, each ~400+ posts) and 10 languages fairly evenly split (13–17 posts/user-language band), indicating the platform serves a genuinely global, non-concentrated user base.

Top engaged users: A small set of power users (e.g. user_zqv2vrf5 at ~90.8K cumulative engagement) drive outsized totals — worth a closer look for bot-like or influencer behavior in later rounds.



4. Deliverables

- Social_Engine_Posts_Cleaned.csv — 12,000 rows, cleaned & standardized
- Social_Engine_Users_Cleaned.csv — 1,500 rows, standardized
- This report